

Role-Aware Multimodal Evidence Decomposition with Language-Anchored Conflict Adjudication for Sentiment Analysis

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Abstract

Multimodal sentiment analysis (MSA) commonly treats textual, acoustic, and visual signals as features to be fused, although these signals do not always play the same predictive role. Some cues express cross-modal agreement, some complement an otherwise reliable interpretation, and others contradict it. Existing shared-specific decomposition and conflict-aware fusion methods only partially expose these roles: residual information is often fused uniformly, suppressed as noise, or handled at the sample level. We introduce CRED-CCDR, a role-aware framework that organizes multimodal evidence into *consensus*, *support*, and *conflict*. A consensus-residual effect-guided disentangler (CRED) first estimates other-modality donor consensus and routes the remaining shared deviations and modality-specific information using both a learned gate and measured cross-modal agreement. A language-anchored conflict adjudication module (CCDR) then interrogates non-verbal conflicts with paired supportive and reversing evidence queries. The final prediction is explicitly decomposed as a consensus estimate plus gated support and conflict corrections. This formulation permits direct analysis of what evidence changes a prediction and in which direction. We specify an evaluation protocol on CMU-MOSI and CMU-MOSEI, including conflict-stratified analysis, ablations, and routing diagnostics. Empirical performance and the semantic interpretation of the evidence roles remain to be established by completed experiments.

Keywords: multimodal sentiment analysis; evidence decomposition; cross-modal conflict; representation disentanglement; language-guided attention

Reviewed working draft (17 September 2026). Experimental results and the corrected architecture figure are pending. The local PDF uses the article fallback when the Elsevier class is unavailable.

1 Introduction

Human sentiment is expressed through words, prosody, facial behavior, and their temporal interaction. Multimodal sentiment analysis (MSA) therefore integrates language, acoustic, and visual observations to estimate affective polarity or intensity. Large benchmark corpora such as CMU-MOSI [1] and CMU-MOSEI [2] have driven rapid progress from tensor fusion [3] and low-rank fusion [4] to cross-modal Transformers [5] and language-centered pretrained models [6]. Yet multimodal communication is not synonymous with uniform multimodal agreement. A positive phrase accompanied by a smile and excited prosody is straightforward; the same phrase uttered with flat prosody or a contemptuous expression demands a different inference process.

Most architectures organize information according to *where it comes from*. Cross-modal attention models exchange features between modality pairs, whereas disentangled methods separate modality-invariant and modality-specific subspaces [7, 8]. This source-oriented view is valuable, but it leaves a second question under-specified: *what role does a piece of information play in the final decision?* Modality-specific content can reinforce an agreed interpretation, oppose it, or be irrelevant. Treating these outcomes as one residual representation forces the final fusion head to infer their roles implicitly.

Recent work has begun to address inconsistency explicitly. ALMT suppresses irrelevant or conflicting non-verbal cues under language guidance [9]. DLF disentangles shared and specific features and uses a language-focused attractor to enhance language representations [8]. TAMSA adopts text-centric asymmetric fine-tuning [10]. Most directly, ECA quantifies emotional conflict, separates similar and dissimilar components, and adapts processing for emotionally difficult samples [11]. These advances establish both the importance of language and the harm caused by unstructured conflicts. However, suppressing, separating, or sample-wise adapting conflict does not answer how each evidence type should affect the regression output. Conflict can be informative rather than simply noisy: sarcasm, hesitation, and mixed affect often appear precisely as disagreement between verbal and non-verbal behavior.

We propose a role-aware alternative. The central hypothesis is that multimodal evidence should be modeled not only by modality source or sharedness, but also by its function in prediction. We distinguish three roles: (i) *consensus* evidence that is supported across modalities; (ii) *support* evidence that complements the consensus; and (iii) *conflict* evidence that may reverse or attenuate it. This distinction leads to an explicit additive prediction:

$$\hat{y} = \hat{y}_{\text{con}} + \Delta_{\text{sup}} + \Delta_{\text{conf}}, \quad (1)$$

where each raw correction is multiplied by a learned scale in $[0, 1]$. The scales attenuate the raw corrections but do not impose an absolute bound on them. Unlike a generic fusion head $f(T, A, V)$, Eq. (1) exposes a base interpretation and the two types of evidence that modify it.

Our framework, CRED-CCDR, is implemented serially after a DLF-style shared-specific representation backbone. CRED computes a other-modality donor consensus for every modality, combines modality-specific content with deviations from that consensus, and routes the resulting residual at the feature level. Its routing gate averages a learned decision with an agreement signal derived from cosine similarity, providing an observable inductive bias without making the route fully deterministic. CCDR then forms a language anchor from linguistic consensus and support. Two learnable queries—one aligned with the anchor and one sign-reversed—attend only to acoustic and visual conflict evidence, enabling the model to search for support and reversal separately. Auxiliary objectives supervise shared representations, encourage separation among evidence roles, and align routing with detached inter-modal predictive disagreement.

The contributions are threefold:

1. We formulate MSA as *evidence-role decomposition*. Beyond shared-specific factorization, multimodal affective cues are organized into consensus, supportive, and conflicting components according to how they influence prediction.
2. We develop CRED and a language-anchored CCDR mechanism. CRED couples other-modality donor consensus with agreement-guided residual routing, while CCDR adjudicates non-verbal conflicts through paired supportive/reversing queries. Their output admits the interpretable decomposition in Eq. (1).
3. We design a conflict-oriented evaluation protocol that complements aggregate benchmark scores with conflict-stratified results, anchor-modality comparisons, correction-effect analysis, routing diagnostics, multi-seed significance tests, and matched-capacity ablations.

2 Related work

2.1 Multimodal sentiment fusion

Early MSA systems combine language, acoustic, and visual cues through feature concatenation or structured multilinear interactions. Tensor Fusion Network (TFN) explicitly models unimodal, bimodal, and trimodal interactions through an outer product [3]; Low-rank Multimodal Fusion (LMF) reduces its computational burden with low-rank factors [4]. MulT replaces explicit

Table 1: Conceptual comparison with representative MSA paradigms. “Explicit effect” indicates that the final score exposes the signed contribution of an evidence role.

Method	Shared/private	Language anchor	Explicit conflict	Support vs. conflict	Explicit effect
MISA [7]	✓	–	–	–	–
ALMT [9]	–	✓	suppression	–	–
DLF [8]	✓	✓	implicit	–	–
TAMSA [10]	–	✓	implicit	–	–
ECA [11]	component-wise	–	✓	–	–
CRED-CCDR (ours)		✓	✓	✓	✓

alignment with directional cross-modal attention and captures long-range dependencies among unaligned sequences [5]. MAG injects acoustic and visual shifts into pretrained language models [6]. MMIM maximizes mutual information hierarchically to retain task-relevant multimodal content [12]. These methods substantially improve fusion, but the fused representation generally does not specify whether a non-verbal cue confirms or contradicts the dominant interpretation.

2.2 Shared-specific representation learning

Heterogeneous modalities contain both common and private factors. MISA projects each modality into modality-invariant and modality-specific subspaces before fusion [7]. Self-MM uses self-supervised unimodal targets to learn differentiated modality representations [13], while TFR-Net reconstructs missing or noisy features for robust prediction [14]. DLF further combines shared-specific disentanglement with a language-focused attractor and hierarchical prediction [8]. Our work adopts the shared-specific separation as a representation backbone, but addresses a semantically different axis: after identifying shared and private information, it asks whether the remaining evidence is consensual, supportive, or conflicting. Thus, “shared versus specific” describes origin and cross-modal commonality, whereas “consensus versus support versus conflict” describes predictive role.

2.3 Language guidance and cross-modal conflict

Language is often the most semantically explicit modality in video-based MSA. ALMT learns an adaptive hyper-modality under multiscale language guidance to suppress irrelevant or conflicting audio-visual content [9]. DLF attracts complementary private information toward language [8], and TAMSA applies asymmetric fine-tuning around a text-centric representation [10]. These studies motivate a language anchor, but language should not be treated as infallible: sarcasm and mixed affect require non-verbal evidence to correct a literal reading.

ECA is the closest conflict-oriented study [11]. It quantifies emotional conflict, separates similar and dissimilar components, and adapts learning for consistent and conflicting samples. CRED-CCDR differs in three respects. First, it performs feature-level role routing rather than only sample-level conflict adaptation. Second, it retains a distinct support path instead of treating all non-consensus content as a single dissimilar component. Third, it parameterizes conflict as a signed correction to a consensus prediction and explicitly evaluates whether that correction improves per-sample error. Table 1 summarizes the distinction; it is not intended as a performance comparison.

3 Method

3.1 Problem definition and overview

For each utterance, the input contains aligned language, acoustic, and visual sequences $X_m \in \mathbb{R}^{T \times d_m}$, where $m \in \mathcal{M} = \{l, a, v\}$. The target $y \in [-3, 3]$ is a continuous sentiment score. The

Architecture figure pending correction.

The supplied visual draft has not been inserted because its encoder, consensus, and conflict-effect paths differ from the implementation.

Figure 1: Reserved architecture figure. The final illustration must follow the equations in Section 3; this box is a review placeholder, not a submission figure.

model predicts $\hat{y} \in \mathbb{R}$ and is trained primarily with an L_1 regression loss. The computation is specified below; Figure 1 reserves space for a code-verified architecture illustration.

The processing is strictly serial. First, a shared-specific backbone produces (C_m, S_m) for every modality. Second, CRED maps these features to consensus Q_m , support U_m , and conflict K_m . Third, pooled consensus produces a base score, pooled support produces a gated support correction, and language-anchored attention over K_a and K_v produces a gated conflict correction. The shared-modality prediction heads are used only for training and routing supervision; they are not averaged into the final output.

3.2 Shared-specific representation backbone

Language tokens are encoded by BERT-base-uncased [15]. Acoustic and visual inputs use the aligned features supplied by the benchmark preprocessing. Each modality is projected into a common d -dimensional space by a temporal convolution:

$$Z_m = \text{Conv1D}_m(X_m), \quad m \in \{l, a, v\}. \quad (2)$$

Padding is masked before and after the convolution. In the MOSI configuration, $T = 50$, the kernel size is 5, and the projected length is 46.

Three modality-specific Transformer encoders and one weight-shared Transformer encoder produce

$$S_m = E_m^s(Z_m), \quad C_m = E^c(Z_m). \quad (3)$$

The shared encoder E^c is reused for all modalities, encouraging a common mapping, whereas E_m^s captures modality-private structure. A lightweight decoder reconstructs the projected input,

$$\tilde{Z}_m = D_m(S_m \parallel C_m), \quad (4)$$

and shared/private separation is encouraged by reconstruction and squared-cosine penalties. This backbone follows the motivation of DLF [8]; our contribution begins with the subsequent role-aware decomposition and replaces DLF’s original language-focused attractor and hierarchical prediction path.

3.3 Consensus-residual effect-guided disentanglement

3.3.1 Other-modality donor consensus

CRED first applies separate, modality-dependent layer normalizations to shared and specific inputs. Within this subsection, C_m and S_m denote these normalized tensors; backbone losses use the encoder outputs before these normalizations. For a target modality m and donor $n \neq m$, cross-modal attention extracts donor content relevant to the target:

$$D_{m \leftarrow n} = \text{MHA}(C_m, C_n, C_n). \quad (5)$$

At every time step, donor reliability is estimated by cosine similarity. With temperature τ ,

$$s_{m,n} = \cos(C_m, D_{m \leftarrow n}), \quad (6)$$

$$\alpha_{m,n} = \frac{\exp(s_{m,n}/\tau)}{\sum_{j \neq m} \exp(s_{m,j}/\tau)}. \quad (7)$$

The modality-conditioned consensus is

$$Q_m = \text{LN} \left(C_m + \sum_{n \neq m} \alpha_{m,n} D_{m \leftarrow n} \right). \quad (8)$$

Only the donor set excludes the target modality: Q_m also retains the target through the residual addition of C_m . It is therefore a target-conditioned consensus candidate, not a target-free estimate or a guarantee of affective agreement.

3.3.2 Residual construction

We remove the estimated consensus from the shared representation and combine the deviation with the modality-specific representation:

$$E_m = C_m - Q_m, \quad (9)$$

$$R_m = P_m(S_m \parallel E_m), \quad (10)$$

where P_m is a two-layer projection with ReLU and dropout. R_m therefore contains both private content and shared information not explained by cross-modal consensus.

3.3.3 Agreement-guided routing

A learned feature-wise gate is computed as

$$G_m^{\text{learn}} = \sigma(H_m(R_m \parallel Q_m)/\tau). \quad (11)$$

To expose cross-modal agreement directly, we also define

$$a_m = \sum_{n \neq m} \alpha_{m,n} s_{m,n}, \quad (12)$$

$$G_m^{\text{agr}} = \frac{\text{clip}(a_m, -1, 1) + 1}{2}. \quad (13)$$

The final gate averages learned routing and measured agreement:

$$G_m = \frac{1}{2} (G_m^{\text{learn}} + G_m^{\text{agr}}). \quad (14)$$

CRED then produces

$$U_m = G_m \odot R_m, \quad K_m = (1 - G_m) \odot R_m, \quad (15)$$

where U_m is support and K_m is conflict. The decomposition is soft: ambiguous features can contribute to both paths. Because $U_m + K_m = R_m$, routing changes role allocation without discarding residual evidence.

3.4 Language-anchored conflict adjudication

3.4.1 Modality-balanced pooling

For a collection of modality sequences F_m , an attention scorer produces temporal weights $w_{m,t}$ and a per-modality summary. The pooled result averages modality summaries rather than concatenating all tokens:

$$\text{Pool}(\{F_m\}_{m \in \mathcal{I}}) = \frac{1}{|\mathcal{I}|} \sum_{m \in \mathcal{I}} \sum_t w_{m,t} F_{m,t}. \quad (16)$$

The set $\mathcal{I}$ contains the modalities supplied to each pool. Separate scorers are used for consensus, support, conflict, and language pooling. The two language pooling calls share parameters but recompute temporal weights for their respective inputs. We obtain global consensus $\bar{Q} = \text{Pool}(\{Q_l, Q_a, Q_v\})$, support $\bar{U}$, conflict $\bar{K}$, and language consensus $Q_l^* = \text{Pool}(\{Q_l\})$.

3.4.2 Consensus and support effects

The base prediction is

$$\hat{y}_{\text{con}} = P_{\text{con}}(\bar{Q} \parallel Q_l^*). \quad (17)$$

Support predicts a raw correction and a reliability scale:

$$\tilde{\Delta}_{\text{sup}} = P_{\text{sup}}(\bar{Q} \parallel \bar{U}), \quad (18)$$

$$\beta_{\text{sup}} = \sigma(F_{\text{sup}}(\bar{Q} \parallel \bar{U})), \quad (19)$$

$$\Delta_{\text{sup}} = \beta_{\text{sup}} \tilde{\Delta}_{\text{sup}}. \quad (20)$$

The scale $\beta_{\text{sup}} \in [0, 1]$ attenuates the raw correction. Since the raw head is unconstrained, neither the correction nor the final prediction has a guaranteed absolute bound. This scale is not a calibrated probability of evidence reliability.

3.4.3 Paired evidence queries for conflict

The language anchor combines linguistic consensus and supportive evidence:

$$H_l = \text{Pool}(\{Q_l + U_l\}), \quad A = \text{LN}(H_l + \bar{Q}). \quad (21)$$

Two learnable slots p^+ and p^- form opposed queries:

$$q^+ = p^+ + A, \quad q^- = p^- - A. \quad (22)$$

They attend to non-verbal conflict tokens only:

$$\begin{aligned} Q_e &= [q^+; q^-], \quad K_{av} = [K_a; K_v], \\ (E^+, E^-) &= \text{LN}_e(\text{MHA}(Q_e, K_{av}, K_{av}) + Q_e). \end{aligned} \quad (23)$$

The opposed anchor offsets bias the two queries toward different evidence. They do not guarantee supportive and reversing semantics; these interpretations require empirical validation. The anchor normalization uses no learned affine parameters, whereas the evidence normalization is a learned LayerNorm. The raw conflict effect and its scale are

$$\tilde{\Delta}_{\text{conf}} = P_{\text{adj}}(\bar{Q} \parallel H_l \parallel E^+ \parallel E^-), \quad (24)$$

$$\beta_{\text{conf}} = \sigma(F_{\text{conf}}(\bar{Q} \parallel \bar{K})), \quad (25)$$

$$\Delta_{\text{conf}} = \beta_{\text{conf}} \tilde{\Delta}_{\text{conf}}. \quad (26)$$

Finally,

$$\hat{y} = \hat{y}_{\text{con}} + \Delta_{\text{sup}} + \Delta_{\text{conf}}. \quad (27)$$

This additive form does not by itself guarantee semantic faithfulness; the training constraints and diagnostics below are therefore integral to the proposed framework.

3.5 Training objectives

Task supervision. The primary objective is $\mathcal{L}_{\text{final}} = |y - \hat{y}|$. Training-only heads P_m^{sh} predict sentiment from each shared representation, and the consensus head is supervised directly:

$$\mathcal{L}_{\text{task}} = \mathcal{L}_{\text{final}} + \lambda_{\text{sh}} \sum_m |y - P_m^{\text{sh}}(C_m)| + \lambda_{\text{con}} |y - \hat{y}_{\text{con}}|. \quad (28)$$

Backbone separation. We use masked input reconstruction and shared/private orthogonality:

$$\mathcal{L}_{\text{rec}} = \sum_m \text{mean}_{\text{valid } t,j} (\tilde{Z}_{m,t,j} - Z_{m,t,j})^2, \quad (29)$$

$$\mathcal{L}_{\text{orth}} = \sum_m \mathbb{E}_t [\cos^2(S_{m,t}, C_{m,t})]. \quad (30)$$

Role decorrelation. To discourage collapsed evidence roles, covariance between each pair of Q_m , U_m , and K_m is penalized:

$$\mathcal{L}_{\text{dec}} = \sum_m \sum_{(A,B) \in \mathcal{P}} \frac{1}{d^2} \|\text{Cov}(A_m, B_m)\|_F^2, \quad (31)$$

where $\mathcal{P} = \{(Q, U), (Q, K), (U, K)\}$. Covariance is computed over valid tokens in the minibatch, centered and divided by $\max(N - 1, 1)$. All task losses are averaged over the minibatch.

Weak routing supervision. The detached shared-head predictions provide a training-only disagreement target. Let $z_m = P_m^{\text{sh}}(C_m)$, $\bar{z} = \frac{1}{3} \sum_m z_m$, and

$$d_m = |z_m - \bar{z}| + \mathbb{I}[\text{sign}(z_m) \neq \text{sign}(\bar{z})], \quad r_m = \frac{d_m}{1 + d_m}. \quad (32)$$

The routed conflict ratio ρ_m is the masked mean of $1 - G_m$. We minimize

$$\mathcal{L}_{\text{route}} = \frac{1}{3} \sum_m (\rho_m - \text{sg}(r_m))^2 + \mathcal{L}_{\text{var}}, \quad (33)$$

Here $\mathcal{L}_{\text{var}} = \mathbb{E}_{i,m} \max(0, 0.01r_{i,m} - \text{Var}_{t \text{ valid}}(\text{mean}_j G_{i,m,t,j}))$, using population variance. It discourages, but does not prevent, constant temporal routing when disagreement is high. Stop-gradient avoids a shortcut in which the shared heads alter their outputs only to reduce routing loss.

Correction direction and support calibration. For non-neutral consensus predictions, support should not reverse the current tendency, whereas conflict should not merely duplicate support. With $d = \text{sign}(\hat{y}_{\text{con}})$,

$$\mathcal{L}_{\text{effect}} = \mathbb{E}[\bar{r} \max(0, d\Delta_{\text{conf}}) + \max(0, -d\Delta_{\text{sup}})], \quad (34)$$

Here $\bar{r}$ is mean detached routing target, and the expectation is restricted to samples with $|\text{sg}(\hat{y}_{\text{con}})| > 0.1$; it is zero if there are none. The direction d is detached. These are soft penalties, not hard sign constraints. On low-conflict samples, a support-only objective encourages $\hat{y}_{\text{con}} + \Delta_{\text{sup}}$ to approach y . The support scale is additionally regressed toward the clipped one-dimensional optimum

$$\beta_{\text{sup}}^* = \text{clip}\left(\frac{y - \hat{y}_{\text{con}}}{\tilde{\Delta}_{\text{sup}}}, 0, 1\right) \quad (35)$$

for $|\tilde{\Delta}_{\text{sup}}| \geq 10^{-4}$. The target is detached, and the gate loss uses Smooth L1 over eligible samples (zero when none qualify). The support-only loss is

$$\mathcal{L}_{\text{sup}} = \frac{\sum_i (1 - \bar{r}_i) |\text{sg}(\hat{y}_{\text{con},i}) + \Delta_{\text{sup},i} - y_i|}{\max(\sum_i (1 - \bar{r}_i), 1)}. \quad (36)$$

The current configuration disables specific-cycle, shared-alignment, role-reconstruction, ranking, and low-conflict correction-magnitude losses. Their presence in the source does not make them active objectives.

The total active objective is

$$\begin{aligned} \mathcal{L} = & \mathcal{L}_{\text{task}} + \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_{\text{orth}} \mathcal{L}_{\text{orth}} + \lambda_{\text{dec}} \mathcal{L}_{\text{dec}} \\ & + \lambda_{\text{route}} \mathcal{L}_{\text{route}} + \lambda_{\text{effect}} \mathcal{L}_{\text{effect}} + \lambda_{\text{sup}} \mathcal{L}_{\text{sup}} + \lambda_{\text{gate}} \mathcal{L}_{\text{gate}}. \end{aligned} \quad (37)$$

4 Experiments

Status: planned evaluation. The following protocols and result tables are proposals, not completed experiments. Dataset files, checkpoints, and per-sample predictions were not supplied for this review.

4.1 Datasets and metrics

CMU-MOSI. CMU-MOSI contains 2,199 opinion utterances from online videos, annotated with sentiment intensity from -3 to 3 [1]. We use the standard speaker-independent splits and the aligned length-50 features distributed with the DLF preprocessing.

CMU-MOSEI. CMU-MOSEI is a larger in-the-wild corpus with more than 23,000 annotated utterances [2]. We use the standard split and the same sentiment regression protocol.

Following prior work, we report mean absolute error (MAE; lower is better), Pearson correlation (Corr; higher is better), seven-class accuracy (Acc-7), and binary accuracy/F1 after excluding zero-labeled samples. Results should be reported as mean $\pm$ standard deviation over five seeds (1111–1115). Statistical comparisons use paired bootstrap resampling over test utterances with 10,000 resamples. **[AUTHOR NOTE: The current train.py invokes only seed 1111; run.py supports the five listed seeds. Populate tables only after the five-seed runs are complete.]**

4.2 Baselines

We compare against representative interaction, representation, language-guided, and conflict-aware methods: TFN [3], LMF [4], MulT [5], MAG-BERT [6], MISA [7], Self-MM [13], MMIM [12], TFR-Net [14], ALMT [9], DLF [8], TAMSAs [10], and ECA [11]. For the most important comparisons (DLF, TAMSAs, and ECA), results should be reproduced under the same feature pipeline where code and data permit. Published numbers may be shown separately but should not be mixed with reproduced numbers without explicit labeling.

4.3 Implementation details

Text is initialized from BERT-base-uncased and fine-tuned with learning rate 2×10^{-5} . All other parameters use AdamW with learning rate 10^{-4} . The common hidden dimension is 50 with 10 attention heads and two Transformer layers. For MOSI, the temporal convolution kernel is 5, batch size is 16, route temperature is 1.5, and CCDD dropout is 0.3. For MOSEI, the kernel is 3 and batch size is 32. The configured active loss weights are

	λ_{sh}	λ_{con}	λ_{rec}	λ_{orth}	λ_{dec}	λ_{route}	λ_{effect}	λ_{sup}	λ_{gate}
value	.05	.20	.05	.01	.01	.10	.01	.05	.05

Table 2: Main results on CMU-MOSI. $\downarrow/\uparrow$ denote lower/higher is better. Values are mean $\pm$ standard deviation over five seeds.

Method	MAE $\downarrow$	Corr $\uparrow$	Acc-7 $\uparrow$	Acc-2 $\uparrow$	F1 $\uparrow$	Source
TFN	TBD	TBD	TBD	TBD	TBD	published/reproduced
LMF	TBD	TBD	TBD	TBD	TBD	published/reproduced
MuT	TBD	TBD	TBD	TBD	TBD	published/reproduced
MISA	TBD	TBD	TBD	TBD	TBD	published/reproduced
Self-MM	TBD	TBD	TBD	TBD	TBD	published/reproduced
MMIM	TBD	TBD	TBD	TBD	TBD	published/reproduced
ALMT	TBD	TBD	TBD	TBD	TBD	published/reproduced
DLF	TBD	TBD	TBD	TBD	TBD	reproduced
TAMSA	TBD	TBD	TBD	TBD	TBD	reproduced
ECA	TBD	TBD	TBD	TBD	TBD	reproduced
CRED-CCDR	TBD	TBD	TBD	TBD	TBD	ours

Table 3: Main results on CMU-MOSEI. Reporting conventions follow Table 2.

Method	MAE $\downarrow$	Corr $\uparrow$	Acc-7 $\uparrow$	Acc-2 $\uparrow$	F1 $\uparrow$	Source
MuT	TBD	TBD	TBD	TBD	TBD	published/reproduced
MISA	TBD	TBD	TBD	TBD	TBD	published/reproduced
Self-MM	TBD	TBD	TBD	TBD	TBD	published/reproduced
MMIM	TBD	TBD	TBD	TBD	TBD	published/reproduced
TFR-Net	TBD	TBD	TBD	TBD	TBD	published/reproduced
ALMT	TBD	TBD	TBD	TBD	TBD	published/reproduced
DLF	TBD	TBD	TBD	TBD	TBD	reproduced
TAMSA	TBD	TBD	TBD	TBD	TBD	reproduced
ECA	TBD	TBD	TBD	TBD	TBD	reproduced
CRED-CCDR	TBD	TBD	TBD	TBD	TBD	ours

Gradient accumulation uses 10 minibatches for MOSI and 5 for MOSEI, giving nominal effective batch sizes of 160 for full accumulation groups. Weight decay is 0.005 and 0.001, respectively. The backbone uses the configured attention mask. Gradient values are clipped at 0.6. The learning-rate scheduler uses ReduceLROnPlateau with factor 0.5 and patience 5, and training terminates after 30 non-improving epochs. Checkpoint selection uses validation loss. Test metrics are also logged each epoch in the supplied trainer; they must not be used for model or hyperparameter selection. The current CSV writer multiplies every metric by 100, including MAE and correlation; restore their intended units before reporting them.

4.4 Main results

Tables 2 and 3 are deliberately left unpopulated until the final five-seed evaluation. Boldface should mark the best result and underlining the second best; significance markers should be used only for reproduced runs with paired predictions.

4.5 Conflict-stratified evaluation

Aggregate scores can conceal whether a conflict-aware model solves the problem it claims to address. We therefore partition each test set into low-, medium-, and high-conflict strata using a *model-independent reference score*. Three unimodal estimators are trained on the training split only, frozen, and applied to the test split. For utterance i , the reference conflict score is

$$c_i = \frac{1}{3} \sum_m |u_{i,m} - \bar{u}_i| + \frac{1}{3} \sum_m \mathbb{I}[\text{sign}(u_{i,m}) \neq \text{sign}(\bar{u}_i)], \quad (38)$$

Table 4: Conflict-stratified MAE on CMU-MOSI. The central claim predicts a larger advantage in the high-conflict subset without degradation on low-conflict samples.

Method	Low	Medium	High	Overall
DLF	TBD	TBD	TBD	TBD
TAMSA	TBD	TBD	TBD	TBD
ECA	TBD	TBD	TBD	TBD
CRED-CCDR	TBD	TBD	TBD	TBD

Table 5: Ablation study on CMU-MOSI. Include parameter-matched controls to separate architectural benefit from parameter count.

Variant	MAE↓	Corr↑	Acc-2↑	High-conflict MAE↓	Params (M)
Full CRED-CCDR	TBD	TBD	TBD	TBD	TBD
Consensus only	TBD	TBD	TBD	TBD	TBD
w/o support effect	TBD	TBD	TBD	TBD	TBD
w/o conflict adjudication	TBD	TBD	TBD	TBD	TBD
Learned routing only	TBD	TBD	TBD	TBD	TBD
Agreement routing only	TBD	TBD	TBD	TBD	TBD
Single conflict query	TBD	TBD	TBD	TBD	TBD
Unit correction scales	TBD	TBD	TBD	TBD	TBD
Parameter-matched MLP control	TBD	TBD	TBD	TBD	TBD

where $u_{i,m}$ is a unimodal prediction and $\bar{u}_i$ their mean. The 1/3 and 2/3 quantile thresholds are fixed from the validation split to avoid adapting the test partition. Test strata need not have equal sizes; report their counts and the rule for tied scores. The reference is independent of the compared fusion models but remains a prediction-based proxy, not annotated ground-truth conflict. The same sample partition is then used for every compared model.

In addition to MAE, we report $\Delta\text{MAE} = \text{MAE}_{\text{baseline}} - \text{MAE}_{\text{CRED-CCDR}}$ per stratum and bootstrap confidence intervals. A convincing result should show that gains increase with c_i ; otherwise the conflict-specific claim must be weakened even if overall accuracy improves.

4.6 Ablation studies

The ablation suite is designed around claims rather than module count:

- **Consensus only:** set both corrections to zero.
- **No support effect:** set $\Delta_{\text{sup}} = 0$.
- **No conflict adjudication:** set $\Delta_{\text{conf}} = 0$.
- **Scalar residual gate:** replace feature-wise G_m with one sample-wise scalar gate.
- **Learned routing only:** remove G_m^{agr} .
- **Agreement routing only:** remove G_m^{learn} .
- **Single conflict query:** replace (q^+, q^-) with one query.
- **Unit correction scales:** set $\beta_{\text{sup}} = \beta_{\text{conf}} = 1$.
- **No role losses:** remove $\mathcal{L}_{\text{dec}}$, $\mathcal{L}_{\text{route}}$, and $\mathcal{L}_{\text{effect}}$.

Table 6: Anchor-modality comparison on CMU-MOSI.

Anchor	MAE↓	Corr↑	High-conflict MAE↓
Language	TBD	TBD	TBD
Audio	TBD	TBD	TBD
Vision	TBD	TBD	TBD
Symmetric multimodal	TBD	TBD	TBD

Table 7: Correction and routing diagnostics.

Diagnostic	Value
Support helpful rate $\Pr(e_s < e_0)$	TBD
Conflict helpful rate $\Pr(e_f < e_s)$	TBD
Conflict harmful rate $\Pr(e_f > e_s)$	TBD
Mean support error reduction	TBD
Mean conflict error reduction	TBD
Gate-target MAE	TBD
Spearman $\rho(\epsilon, c)$	TBD

4.7 Is language the appropriate anchor?

To test rather than assume linguistic dominance, we repeat adjudication with language, audio, vision, and symmetric multimodal anchors while holding all other components fixed. We report both overall and high-conflict MAE. Language anchoring is supported only if it consistently outperforms alternatives; if a symmetric anchor is comparable, the claim should be revised from language necessity to a practical inductive bias.

4.8 Correction-effect and routing analysis

For each test sample, define the consensus error $e_0 = |y - \hat{y}_{\text{con}}|$, the post-support error $e_s = |y - (\hat{y}_{\text{con}} + \Delta_{\text{sup}})|$, and final error $e_f = |y - \hat{y}|$. We report the rates $\Pr(e_s < e_0)$ and $\Pr(e_f < e_s)$, mean error reduction, harmful-correction rate, overshoot rate, and gate-target MAE. These quantities distinguish a branch that changes predictions from one that changes them usefully.

We additionally test whether conflict energy

$$\epsilon_i = \frac{1}{3} \sum_m \text{mean}_{t,d} K_{i,m,t,d}^2 \quad (39)$$

is monotonically associated with the independent conflict score c_i using Spearman correlation. Feature-role separation is quantified by pairwise covariance norms and linear-probe leakage among Q , U , and K . Attention visualizations are used only as qualitative support, not as a standalone explanation of model behavior.

4.9 Efficiency and robustness

We report trainable parameters, floating-point operations per utterance, peak GPU memory, and inference latency for DLF, ECA, a parameter-matched DLF control, and CRED-CCDR. Robustness is evaluated by masking each auxiliary modality and by adding controlled Gaussian noise to acoustic and visual features. These experiments determine whether the explicit evidence paths improve robustness or merely increase capacity.

5 Discussion

The proposed framework is intentionally more structured than unconstrained fusion. Its benefit is testability: consensus, support, and conflict yield separate representations, signed corrections, and measurable failure modes. This structure also imposes assumptions. First, support is encouraged to preserve the sign of the consensus, which is appropriate for complementary intensity but can be restrictive near neutral sentiment. Second, the language anchor reflects the semantics of English opinion videos and may be less suitable when facial or acoustic behavior is primary. Third, weak routing supervision depends on auxiliary predictive disagreement, which is not ground-truth evidence-role annotation. The independent conflict-stratified protocol is therefore essential to avoid validating routing only against its own training target.

The relationship to ECA deserves particular emphasis. Both methods recognize that cross-modal conflict should not be absorbed silently by a generic fusion block. ECA adapts representation learning to the difficulty of emotionally conflicting samples, whereas CRED-CCDR decomposes each modality’s residual evidence and estimates its signed effect on the output. These approaches are potentially complementary: ECA-style sample difficulty could regularize CRED routing, and role-decomposed corrections could make conflict adaptation more inspectable.

6 Limitations and broader impact

The current study uses benchmark-aligned features and does not establish performance on raw video or out-of-domain speakers. MOSI and MOSEI are English, opinion-centric, and collected from online videos; their demographic and cultural coverage is limited. Sentiment labels are subjective, and cross-modal incongruity can reflect personality, disability, culture, recording artifacts, or annotation noise rather than deception or sarcasm. Accordingly, conflict energy must not be interpreted as a psychological diagnosis or a measure of truthfulness. The method should not be deployed for high-stakes decisions about individuals without domain-specific validation, fairness analysis, and human oversight.

7 Conclusion

We presented CRED-CCDR, a role-aware approach to multimodal sentiment analysis. Starting from shared and modality-specific representations, CRED extracts other-modality donor consensus and routes residual information into support and conflict using learned and similarity-based signals. CCDR then uses paired language-anchored queries to adjudicate non-verbal conflict. The resulting prediction decomposes into a consensus estimate and gated support/conflict effects. The proposed evaluation goes beyond aggregate accuracy by testing whether benefits concentrate on independently identified disagreement samples and whether each correction actually reduces error. Final conclusions regarding empirical superiority will be inserted after the complete five-seed benchmark and ablation suite is finished.

CRedit authorship contribution statement

Anonymous Author(s): Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing–original draft, Writing–review & editing. [AUTHOR NOTE: Replace with the actual author-level CRediT statement before submission.]

Declaration of competing interest

[AUTHOR NOTE: Authors must provide and confirm the competing-interest declaration.]

Data and code availability

CMU-MOSI and CMU-MOSEI are available through the CMU Multimodal SDK subject to their respective terms. An anonymized implementation and experiment configuration will be released at **TBD** upon acceptance. [AUTHOR NOTE: Provide an anonymous repository during review if permitted by the journal.]

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